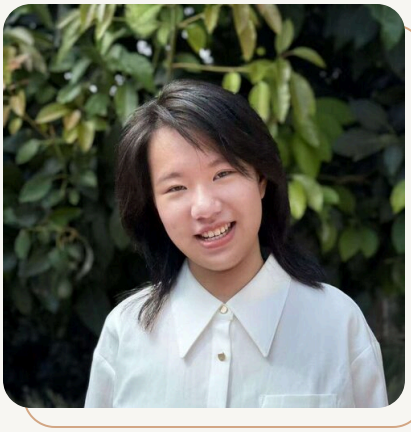
ELECTRICAL & EMBEDDED SYSTEMS ENGINEER

Xiyuan (Amy) Liu

I design, bring up and verify hardware: mixed-signal PCBs, avionics display electronics, FPGA logic and the embedded firmware that ties it together.

Harvey Mudd College · B.S. Engineering (ECE), expected May 2027 · San Jose, CA

xiyuanliu-amy-blog.netlify.app | amyliu01@g.hmc.edu | LinkedIn | GitHub



6
avionics display panels taken to production release

450+
hardware requirements authored in DOORS

35
formal proofs written for a RISC-V quantum controller bridge

1st
place at FAR Unlimited 2026 as rocketry avionics lead

SKILLS

Hardware
PCB schematic & layout, mixed-signal & RF, SI / EMI debugging, bring-up, DO-160G / MIL-STD qualification

Digital & embedded
SystemVerilog, FPGA, embedded C, ARM Cortex-M, RISC-V, UVM & formal verification, JTAG, Python

Tools
Altium, Xpedition, KiCad, LTspice, Questa, Verilator, SymbiYosys, MATLAB / Simulink, ANSYS, DOORS

Lab
Oscilloscope, VNA, logic analyzer, DMM

RECOGNITION

- RTX Display Department Outstanding Employee, Certificate of Excellence (2026)
- Passed FE Electrical & Computer Engineering (EIT), 2026
- 1st, 3rd and 5th at FAR Unlimited (2026, 2025, 2024)
- 3rd place, HMC ICPC qualifier (2022, 2023)

IN THIS PORTFOLIO

- RTX Collins Aerospace: avionics display hardware co-op
- HRL Laboratories: RISC-V quantum controller verification
- Rocketry avionics lead, plus clinic and research work
- Course projects: FPGA/MCU systems, RF, processor design, filters, control

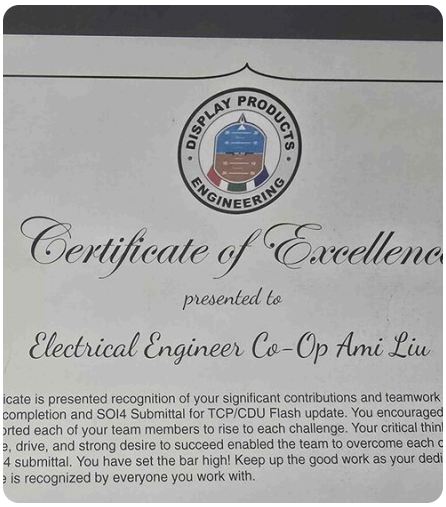
FULL PORTFOLIO ONLINE

Every title in this PDF links to its page, with more photos, plots, video and full lab write-ups.

xiyuanliu-amy-blog.netlify.app

Industry & clinic

all experience →



INDUSTRY

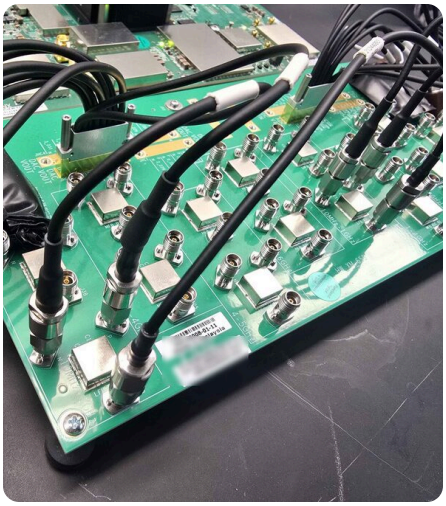
RTX Collins Aerospace

Electrical Engineer Co-Op · June 2025 – August 2026

Hardware design, bring-up, verification and qualification for avionics display control panels. Six panels moved through final production release while I was there.

- Designed a level-shifted JTAG adapter in Altium that released first pass with no re-spin.
- Fixed a DO-160G radiated-emissions failure by damping a high-speed clock, recovering three weeks of schedule.
- Authored 450+ requirements and 300+ test cases in DOORS; reworked a card for proton-radiation testing of a replacement flash.

Read more →



CLINIC · OPEN SOURCE

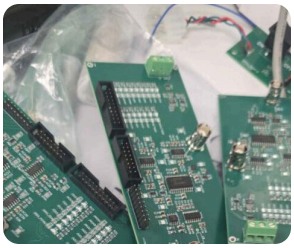
HRL Laboratories

Electrical & Embedded Systems Engineer, Verification (Clinic) · September 2026 – present

Replacing QICK's proprietary timed processor with a RISC-V controller, so quantum control firmware can use standard RISC-V tools. I designed the team's verification plan and built its flow.

- Differential verification against the original processor in a Verilator environment.
- Timing monitors at 7 pipeline checkpoints across 3 clock domains; 322 pulses characterized.
- 35 SymbiYosys formal tasks that exposed three ways pulses could be silently lost.

Read more → Code on GitHub →



CLINIC

U.S. Department of Agriculture × Auburn University

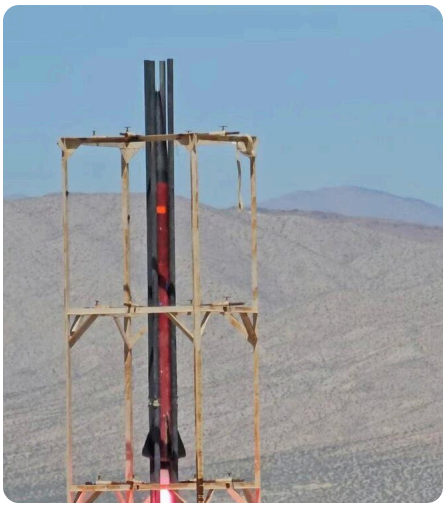
Electrical Engineer (Clinic) · August – December 2024

Worked on the electronics for a sponsored Harvey Mudd Clinic project modernizing electropenetograph technology, an instrument entomologists use to study how insects feed on plants.

Project overview on HMC Clinic →

Rocketry avionics

full page with onboard video →



LEADERSHIP

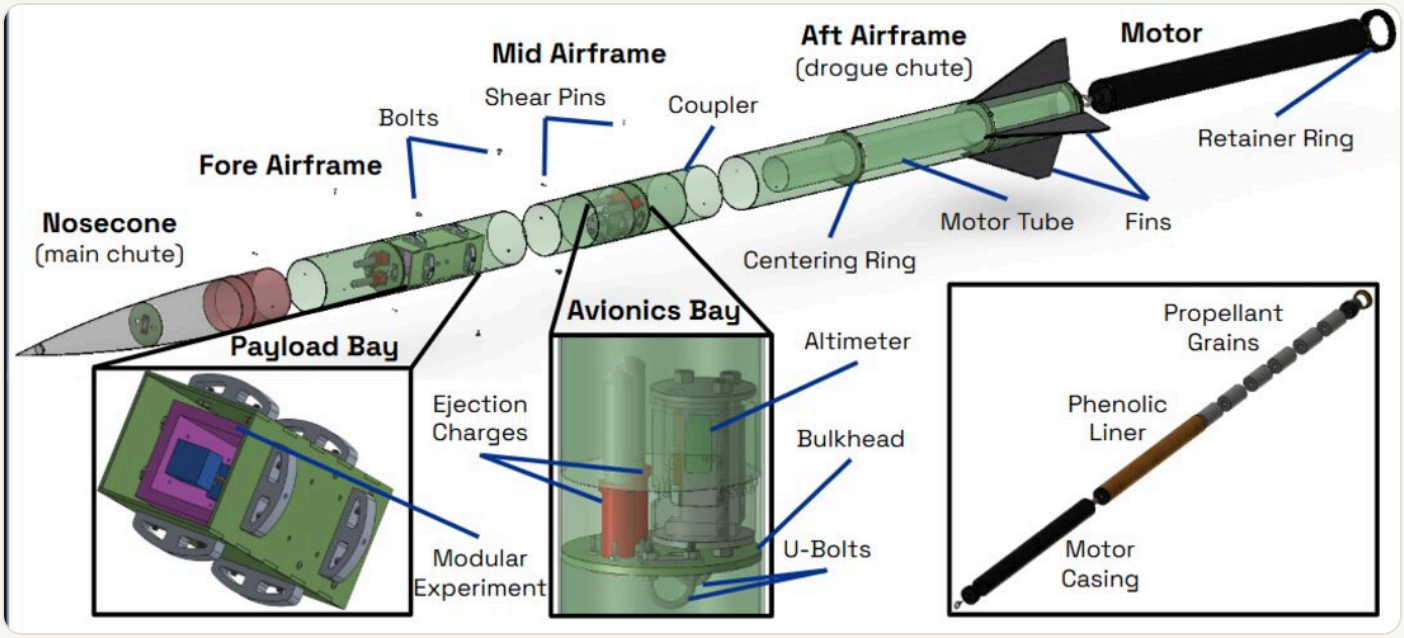
Mudd Amateur Rocketry Club

Avionics Lead · September 2022 – present

Leading avionics for Harvey Mudd's competition rocketry team, which placed 5th, 3rd and then 1st at FAR Unlimited from 2024 to 2026.

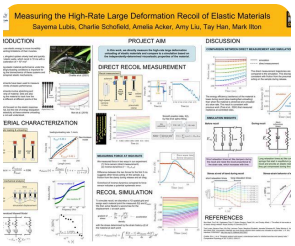
- Combined redundant sensors so the state estimate holds through high-g saturation, cutting apogee-detection timing error by about 30%.
- Building Atlas, a custom STM32H743 flight computer for the team's first actively guided rocket (work in progress).
- Wrote a Level 1 design guide and led build sessions for 60+ students.

Read more → MARC on Instagram →



Dual-deploy layout: main chute in the nosecone, drogue in the aft airframe, avionics bay with altimeter and ejection charges between them.

Research



RESEARCH

Physics of Soft Matter Lab

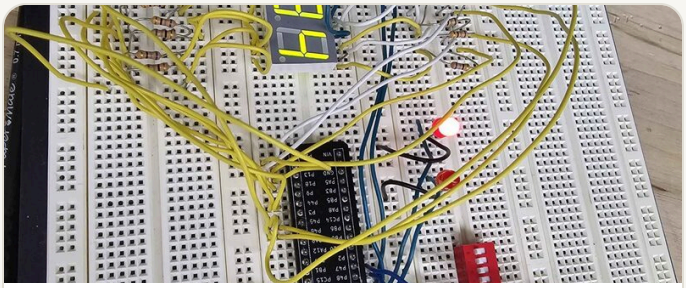
Research Assistant · September 2023 – January 2025

Studied latch-mediated spring-actuated systems and automated the lab's recoil-motion analysis with MATLAB computer vision. Co-authored a poster presented at the SoCal Soft Matter Symposium and the APS Physics Global Summit.

Read more →

Course projects

all projects →

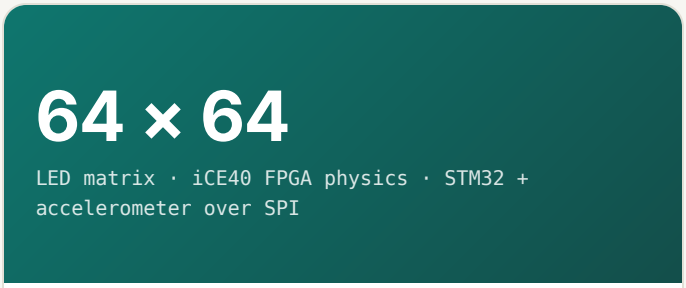


E155 · FALL 2024

Microprocessor Systems

Seven projects on an iCE40 FPGA and an STM32: display multiplexing, keypad scanning, timers, interrupts (0.05% speed error) and an SPI + Wi-Fi temperature server.

Labs 1–6 →

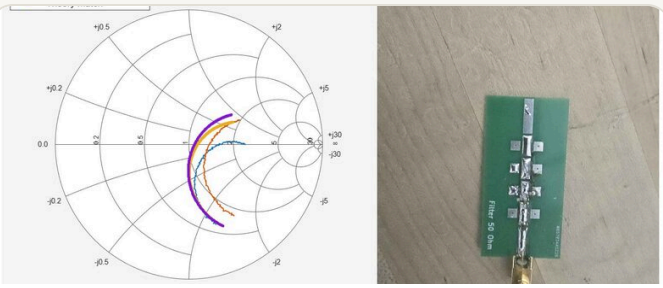


E155 FINAL PROJECT · WITH STEPHEN XU

Interactive 2D Physics Engine

A tilt-controlled sand simulator. The FPGA runs particle physics on a 64×64 LED matrix in 12-bit color while an STM32 streams gravity direction from an accelerometer over SPI.

Project page with demo →Code →

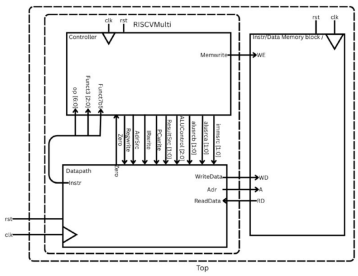


E157 · FALL 2026

RF Circuit Design

Predict, simulate, then measure. Transmission line reflections with extracted parasitics, VNA calibration, and a hand-soldered 270 Ω to 50 Ω L-match at 100 MHz.

Labs →

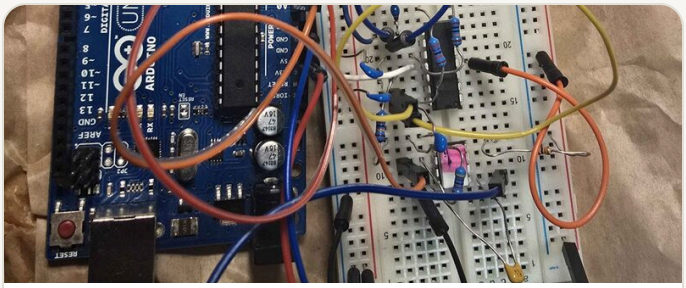


E85

Multicycle RISC-V Processor

Controller and datapath in hierarchical SystemVerilog, passing all 40 controller test vectors and traced cycle by cycle through a test program.

Labs 10–11 →

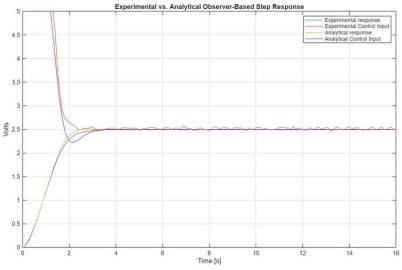


E101 · WITH LERAN HU AND BETSY WU

Analog & Digital Filters

A 4th-order Sallen-Key Butterworth filter plus a comb filter on an Arduino turn a 10 Hz pulse train into a clean sine, checked harmonic by harmonic.

Project →



E102 · SPRING 2025

Digital Controller for an RC Circuit

PI and observer-based LQR controllers designed in MATLAB/Simulink and run on an Arduino at 10 Hz. The observer settled in about 3 s with under 0.4% overshoot.

Project →